Course Information

Course Number: DAEN 460
Course Title: Capstone Senior Design
Section: 900
Time / Location: T 12:45-3:15 PM / ZACH 211
Credit Hours: 3 (1 hour of lecture, 6 hours of lab per week)

Instructor

Instructor: Alexander Abuabara
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Office Hours: W 1-3 PM (or by appointment)

Course Description

DAEN 459

First in a two-course sequence for the capstone senior design experience; formation of a senior design team, visitation with the team sponsor, preparation of the groundwork for the project, preparation of the project charter and collection or acquisition of initial set of data; provision of instructions on different aspects of capstone design, including ethics, design constraints, applicable standards, project management, report writing specifications and requirements, and oral and visual presentations.

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Second course in a two-course sequence for the capstone senior design experience; continuation of work on the senior design project in teams; data collection, analysis, application of data engineering methods and tools and development of recommendations considering design constraints, evaluation of alternative design, and application of relevant standards; engagement in oral presentations and creation of the project report, with relevant feedback provided during the semester.

Pedagogical Model

Semester 1 (DAEN 459) = "Design and De-risk"
Students should:

- Define the problem,
- Acquire data,
- Validate feasibility,
- Design the system,
- Produce operational plans.

Semester 2 (DAEN 460) = "Build and Deploy"
Students should:

- Implement,
- Test,
- Iterate,
- Evaluate,
- Present final system.

Learning Outcomes

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By the end of the course, students will be able to:

- Define and scope a feasible capstone project with clear goals, objectives, and deliverables.
- Develop and validate project requirements in collaboration with the project sponsor and relevant stakeholders.
- Design appropriate system architecture and infrastructure plans to support project implementation.
- Identify, collect, acquire, and prepare the initial datasets required for the project.
- Evaluate data governance, privacy, security, ethical, and responsible-use considerations relevant to the project.
- Develop a realistic project schedule that includes milestones, dependencies, resources, risks, and mitigation strategies.
- Design and develop an initial prototype or proof-of-concept that demonstrates the feasibility of the proposed solution.
- Prepare and deliver a professional project proposal and presentation that clearly communicates the problem, proposed solution, technical approach, implementation plan, and expected outcomes.

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By the end of the course, students will be able to:

- Implement and refine a comprehensive capstone project solution based on the approved project scope, requirements, and design established in DAEN 459.
- Collect, integrate, clean, transform, and manage project datasets using appropriate data engineering methods, tools, and technologies.
- Develop, implement, and evaluate data engineering pipelines, system architectures, and analytical solutions that address the project objectives and sponsor requirements.
- Apply appropriate data analysis, modeling, visualization, and evaluation techniques to generate meaningful insights and support data-driven decision-making.
- Evaluate alternative technical solutions and justify design decisions based on performance, scalability, reliability, cost, and other relevant engineering constraints.
- Apply relevant industry standards and best practices for data governance, data quality, privacy, security, ethics, and responsible data use throughout project implementation.
- Validate and assess the performance, effectiveness, and limitations of the developed solution using appropriate testing methods, evaluation metrics, and stakeholder feedback.
- Collaborate effectively with team members, project sponsors, and relevant stakeholders to manage project progress, address technical challenges, and deliver a completed engineering solution.
- Develop actionable recommendations and identify opportunities for future improvements based on project findings, evaluation results, and sponsor needs.
- Prepare and deliver a comprehensive technical project report, professional presentation, and demonstration that clearly communicate the implemented solution, engineering methods, results, limitations, and recommendations.

Why I Teach this Course

This course offers you a unique learning experience. I am always especially excited to teach this course, here is why:

- **You finally get to combine everything that you have learned throughout your entire undergraduate curriculum**. This not only includes the data engineering skills you have gained, but also the professional, cultural, and communication skills that you have developed.
- **This is an opportunity for you to try to solve a problem that matters to you**. This course gives you a lot of freedom and choice in an extensive project. This course can be the first step towards moving to the career of your choice.
- **Building a system from start to finish is fun and rewarding**. Students often tell me that this was the greatest experience of their entire college career.
- **Working on teams can be messy, but it's worth it**. It is amazing what you can accomplish together, so much more than on your own. Working on real problems with real people has its challenges, but it is also very rewarding when you successfully complete your goals.
- **Every semester is different, every team is different, every solution is different**. There is always something new to learn and innovate.
- **Smaller class sizes and weekly meetings allow you to connect** with your classmates and your instructor(s).

Course Expectations

As a data engineering student, you are expected to apply the skills and knowledge developed throughout your academic career toward the planning and development of a substantial, real-world data-centric project. These expectations mirror professional data engineering roles, preparing you for collaborative, cross-functional work. You are expected to:

- Contribute actively to your team with clear communication, accountability, and collaboration,
- Attend all meetings, meet deadlines, contribute to team decisions, and resolve conflicts professionally,
- Work with stakeholders to define and refine technical and business requirements,
- Develop a detailed project plan with timelines, data needs, and deliverables,
- Address ethical, privacy, and risk considerations,
- Identify, acquire, clean, and validate datasets, including the elaboration of data dictionaries and relevant metadata,
- Design a robust data architecture, including storage, pipelines, and processing frameworks,
- Document data flow, system architecture, and all planning and data engineering decisions with supportive analysis,
- Present project components clearly to both technical and non-technical audiences,

- Deliver a comprehensive design proposal with diagrams, models, and a clear roadmap,
- Track progress through milestone reviews and maintain an updated project board,
- Manage scope, time, and resources, adjusting plans as needed,
- Apply best practices in version control, reproducibility, and testing,
- Ensure compliance with industry standards, regulations, and school policies, and
- End the course with a feasible, well-scoped project ready for implementation.

Recommended/Reference Textbooks

Available free online at TAMU Libraries:

- **Design of Everyday Things**. D. Norman (2013)
- **Universal Principles of Design**. W. Lidwell, K. Holden, J. Butler (2023)
- **Tragic Design: The Impact of Bad Product Design and How to Fix It**. J. Shariat, C. Savard Saucier (2017)
- **Fundamentals of Data Engineering**. J. Reis, M. Housley (2022)
- **Designing Data-Intensive Applications**. M. Kleppmann (2017)
- **The Data Warehouse Toolkit**. R. Kimball, M. Ross (2013)

Grading Policy

- Graded items and weights:

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	Deliverable		**Weight**
1	Professionalism and participation (including peer-review)	Individual	11%
2	Team agreement	Individual	2%
3	Sponsor meeting notes	Individual	5%
4	Problem statement and project objectives	Team	7%
5	Draft project charter	Team	2%
6	Ethics, privacy, and risk assessment	Individual	10%
7	Data requirements and acquisition plan	Team	10%
8	Initial data inventory and profiling report	Team	4%
9	Preliminary system/data architecture	Team	10%
10	Requirements and constraints matrix	Team	7%
11	Work breakdown structure and project schedule	Team	10%
12	Risk register and mitigation plan	Individual	5%
13	Evaluation and verification plan	Team	5%
14	Draft planning report and presentation	Team	1%
15	Final planning report, charter, presentation, and DAEN 460 transition package	Team	11%

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	Deliverable		**Weight**
1	Professionalism and participation (including peer-review)	Individual	15%
2	Presentation 1	Team	15%
3	Presentation 2	Team	10%
4	Presentation 3	Team	10%
5	Presentation 4	Team	20%
6	Final report	Team	30%
7	Writing Assignment 1	Individual	
8	Writing Assignment 2	Individual	
9	Writing Assignment 3	Individual	
10	Writing Assignment 4	Individual	

- In addition to the graded items listed above, other submissions will be evaluated on a **satisfactory/unsatisfactory** basis. Details of these requirements and milestones will be posted on the Canvas course page and may include review sessions as well as meetings with sponsors and advisors to discuss strategies. For example,

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  - Individual skills profile and project preferences
  - Team agreement and preliminary role assignments
  - Sponsor meeting notes and stakeholder register
  - Problem statement and project objectives
  - Draft project charter
  - Ethics, privacy, and risk assessment
  - Data requirements and acquisition plan
  - Initial data inventory and profiling report
  - Preliminary system/data architecture
  - Requirements and constraints matrix
  - Work breakdown structure and project schedule
  - Risk register and mitigation plan
  - Evaluation and verification plan
  - Draft planning report and presentation

  DAEN 460
  - Updated code of conduct
  - Mini project management packet
  - Sponsor meeting log
  - Advisor review meeting log
  - Showcase registration and poster
- Grades will be calculated by totaling the points you earn and comparing them to the max. possible points.
- Grades assigned are A: 90%–100%, B: 80%–89.9%, C: 70%–79.9%, D: 60%–69.9% and F for less than 60%.

Late Work Policy

Due to the pace and structure of this capstone course, late or make-up work cannot be accommodated. Late or missed assignments will receive a zero. University-excused absences apply only to attendance and in-class activities and do not automatically extend assignment deadlines; any alternative arrangements must be made before the deadline. Grade concerns must be raised within one week of the graded work being returned. Regrade requests will result in the entire submission being regraded.

Course Assignments

Initial guidelines are in the appendix below; more detailed instructions will be shared in class.

Professionalism and participation means being responsible, prepared, and actively involved throughout the project. This includes attending meetings, completing assigned tasks on time, communicating clearly with teammates and instructors, accepting feedback, and contributing ideas to solve problems. Professionalism also means producing accurate and organized work, following ethical practices when handling data, and using appropriate technical tools and documentation. Strong participation shows that a student is dependable, works well with others, and contributes consistently to the success of the project.

All submissions and presentations are formal, graded deliverables and must be uploaded to Canvas as assigned. All team members must participate, with leadership rotating across the team. Teams should hold a brief debrief after presentations. The course concludes with a final poster presentation of each team's capstone project plan. All DAEN capstone teams participate.

Teams are formed by the instructor based on preferences or direct project assignment. Each team has an ISEN faculty advisor, with the instructor guiding the course and the sponsor serving as the client.

Faculty Advisor – Each team must have a faculty advisor from the ISEN department, and the instructor can suggest advisors who fit the project. The advisor is your subject matter expert, offering guidance and suggestions along the way. Teams should meet with their advisor every two weeks unless the advisor approves a different schedule. Faculty advisors are actively involved and considered part of your team, so treat them with respect and courtesy.

Additional Faculty Advisor – Teams can add a secondary advisor after checking with the course instructor. These advisors can be from any Texas A&M department or affiliated organization. If you want an advisor from outside Texas A&M, you'll need written approval from the instructor. The request should include their name, email, phone number, organization, area of expertise, and a short explanation of why their help would be valuable.

Course Material and Copyright

All documents used in this course are copyrighted. Here, "documents" means all materials generated for this class, including, but not limited to, course syllabi, course notes, quizzes, exams, problem sets, slide sets, and all materials appearing on the Canvas website or sent to you via email. Because these materials are copyrighted, you do not have the right to copy any of them for any purpose other than your own personal academic use unless the instructor expressly grants permission. Course materials may not be given or sold to any for-profit enterprise.

Confidentiality and Non-attribution

In this course, we will often use client data, project details, or conversations that need to stay within our group. You can share this information with classmates and the teaching team, but not outside the course. To keep things confidential, store notes or files in a safe way (like a folder, binder, or password-protected device) and keep them with you or in a secure place (room, backpack, car). Non-attribution means focusing on ideas, not people. Instead of saying "*José said Range Rovers have reliability issues*," say "*I understand that Range Rovers have reliability issues*." Avoid tying insights to individuals or teams.

Attendance & Class Participation

Because this is a capstone course, regular attendance and active participation are essential to successful completion of the course. Much of the work will involve collaborative activities, project meetings, presentations, peer feedback, and other activities that cannot be fully replicated outside of class.

Students are expected to attend all scheduled class meetings, arrive on time, and remain for the full class period. Students are also expected to attend required team, project, client, or presentation meetings identified by the professor as part of the course.

If an absence is unavoidable, students should notify the professor as soon as reasonably possible and are responsible for obtaining information about missed material, assignments, deadlines, and team responsibilities. An absence does not automatically excuse a student from completing assigned work or meeting course deadlines.

Documented and verifiable absences will be handled in accordance with university policy. When necessary and appropriate, alternative or make-up work may be assigned. Make-up assignments or assessments may be administered in a different format, including orally, upon the student's return, on a date determined by the professor.

Because capstone work often depends on collaboration, students are also responsible for communicating promptly with their project team when an absence affects group responsibilities. Repeated absences, tardiness, early departures, or failure to participate in required course activities may affect the student's participation, professionalism, project, or other applicable course grades.

Students with unusual or extended circumstances affecting attendance should contact the professor as early as possible so that appropriate arrangements can be considered consistent with university and course policies.

Team Meetings – Active engagement is a key part of the capstone experience. Students are expected to fully participate in all team activities, including meetings with sponsors, faculty advisors, and internal discussions. These interactions help the

project succeed while giving you opportunities to collaborate, solve problems, and practice professional communication skills. Meaningful participation benefits both you and your team throughout the course.

In team projects, everyone is expected to contribute fairly. When someone doesn't, the rest of the team carries the burden, which isn't fair. This problem, often called social loafing, can take many forms:

- Ignoring emails, texts, or calls from the team.
- Being told there's a communication issue but not fixing it.
- Expecting the team to work around your personal schedule.
- Missing meetings, showing up late, or leaving early.
- Submitting work that is incomplete, full of mistakes, or unusable.
- Working on unrelated ideas without team approval, then expecting credit.
- Attending meetings but spending more time on your phone than participating.

What's expected of you:

- Be available, stay in touch with your team. It's your responsibility to make sure communication works.
- Do your part! All must contribute a fair share of the project. Fair doesn't always mean equal. Tasks may vary. Teams are encouraged to assign work that fits members' skills, but everyone must contribute in a meaningful way. Participation (or lack of it) will be reflected in presentations and peer reviews.
- Always show courtesy and respect! Professional behavior is always expected. Teammates are colleagues, use respectful language, and contribute in a way that shows maturity and responsibility.
- If a student consistently fails to participate, faculty may move them into a *Track B Option*: working alone under direct supervision. In this track, the highest possible grade is a C (if all requirements are met).
- Speak up early! If a teammate still isn't contributing after a TA discussion, let faculty know right away. Problems don't fix themselves, and waiting until the end of the semester leaves no time for solutions.

Project Sponsor Interaction – Teams should meet with the project sponsor's point of contact (POC) about once a month, adjusting as needed for the sponsor's schedule. Meetings can be by phone or video call (email doesn't count). All team members should attend unless there's a university-excused absence or a class conflict. The team and sponsor will coordinate meeting times, and only the sponsor can cancel. If a meeting is canceled or the sponsor feels the schedule isn't working, note it in your technical presentations and discuss a solution with the instructor.

Courses Schedule (Tentative)

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Week	Topics	Team / Lab activities	Milestones / Deliverables
1	Capstone expectations; design process; professional conduct	Review sponsor projects; skills and interest inventory	Individual skills profile and project preferences
2	Team formation; teamwork; roles and responsibilities	Form teams; select projects; establish communication practices	Team agreement and preliminary role assignments
3	Sponsor engagement; stakeholder analysis; effective interviewing	Initial sponsor meeting; identify stakeholders and decision context	Sponsor meeting notes and stakeholder register
4	Problem framing; needs assessment; requirements elicitation	Translate sponsor needs into a preliminary problem statement	Problem statement and project objectives
5	Project charters; scope definition; assumptions and constraints	Define scope, exclusions, success criteria, and major deliverables	Draft project charter
6	Ethics, privacy, security, bias, and responsible data use	Conduct ethical and data-governance review	Ethics, privacy, and risk assessment
7	Data requirements; data provenance; acquisition strategies	Inventory available data; identify gaps and access dependencies	Data requirements and acquisition plan
8	Initial data collection and profiling	Acquire an initial dataset; assess schema, quality, completeness, and suitability	Initial data inventory and profiling report
9	Data engineering architecture and tool selection	Develop source-to-output architecture; evaluate platforms and tools	Preliminary system/data architecture

10	Applicable standards; legal, organizational, and design constraints	Identify standards and convert constraints into requirements	Requirements and constraints matrix
11	Project management; work breakdown structures; estimation	Develop tasks, dependencies, schedule, ownership, and communication plan	Work breakdown structure and project schedule
12	Risk management; feasibility; change control	Assess technical, data, schedule, personnel, and sponsor risks	Risk register and mitigation plan
13	Evaluation planning; alternatives; metrics and acceptance criteria	Define baseline, alternatives, validation methods, and performance measures	Evaluation and verification plan
14	Technical writing; visual communication; design reviews	Integrate planning artifacts; peer and instructor design review	Draft planning report and presentation
15	Professional presentations; transition to implementation	Formal sponsor presentation; revise plans based on feedback	Final planning report, charter, presentation, and DAEN 460 transition package

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Week	Topics	Team / Lab activities	Milestones / Deliverables
1	Project reactivation; execution expectations; configuration management	Reconfirm scope, roles, sponsor expectations, data access, and schedule	Updated project plan and kickoff memorandum
2	Reproducible development; repositories; testing strategy	Configure development environment, repository workflow, and issue tracking	Reproducibility and testing plan
3	Data ingestion and acquisition	Implement source connections and initial ingestion processes	Working ingestion prototype, PRESENTATION 1
4	Data quality engineering; validation and observability	Implement schema, completeness, accuracy, and anomaly checks	Data-quality rules and baseline assessment, PRESENTATION 1
5	Data transformation; integration; metadata and lineage	Build repeatable cleaning, transformation, and integration workflows	End-to-end pipeline increment
6	Storage, data modeling, performance, and security	Implement storage/modeling layer; review access controls	Architecture and implementation review
7	Exploratory analysis; baselines; analytical validity	Establish descriptive findings and baseline performance	Technical progress report, PRESENTATION 2
8	Midsemester design review	Demonstrate pipeline and preliminary analysis to faculty and sponsor	Working prototype, PRESENTATION 2
9	Application of analytical or data engineering methods	Implement primary models, algorithms, or decision-support methods	Analysis/modeling increment
10	Alternative designs; trade-off analysis; design constraints	Compare approaches using technical and stakeholder criteria	Alternatives and trade-off analysis, PRESENTATION 3
11	Verification, validation, testing, and uncertainty	Perform system tests, sensitivity analysis, error analysis, or model validation	Validation and testing report, PRESENTATION 3
12	Interpretation; recommendations; operational implications	Convert results into sponsor-relevant findings and recommendations	Draft findings and recommendations
13	Deployment, maintainability, documentation, and handoff	Package pipeline, code, dashboard, model, or other product for sponsor use	Release candidate and technical documentation, PRESENTATION 4
14	Report writing; presentation design; peer review	Conduct final technical review and rehearsal; resolve remaining defects	Draft final report, PRESENTATION 4

15	Final design presentation and project handoff	Present results; demonstrate product; transfer artifacts to sponsor	Final report, showcase presentation, product, repository, and sponsor handoff

Tips – (i) Be nice. Be honest. Don't cheat. **Adhere to all University Policies**. (ii) You all are important to me: include the course number in email subject so I can prioritize it. (iii) Every step matter and each build on the next: stay engaged, keep up, ask questions. (iv) Learning takes steady effort: start early, not last-minute cramming. (v) Engineers turn data into meaningful information, practice here, get prepared. (vi) Clear, well-organized work is as important as the content itself. (vii) Use others' ideas to build your own but always credit your sources: good scholarship means giving credit where it's due. (viii) No audio or video recording without permission; it's a conduct violation otherwise.

Technology Support – For technical support contact isen-helpdesk@tamu.edu.

AI Statement

Students are permitted to use artificial intelligence tools as part of their work in this course. AI should be treated like any other academic or professional tool: it may assist with brainstorming, organization, research, revision, or problem-solving, but the student remains responsible for the final work submitted.

Students are responsible for verifying the accuracy, quality, and appropriateness of any information produced with AI. Submitting inaccurate, fabricated, or misleading information generated by an AI tool does not excuse the student from responsibility for the work.

Any ideas, information, quotations, images, or other material drawn from outside sources must be properly cited according to the citation style required for the assignment. When AI has contributed substantially to an assignment, students should also acknowledge or cite their use of the AI tool when appropriate.

AI should support your learning—not replace your own thinking, judgment, or understanding. Students may be asked to explain, discuss, or demonstrate the reasoning behind work they submit. Use AI in a way you can feel confident about and openly acknowledge. If you would be uncomfortable explaining how you used AI, reconsider whether that use is appropriate for the assignment.

Initial guidelines – DAEN 459

This capstone course serves as an integrative, high-impact educational experience that challenges students to integrate their accumulated knowledge and apply it to a complex, real-world problem. The capstone projects encourage students to adopt a hands-on approach by building solutions from the ground up rather than relying only on existing tools or templates. The goal is to improve essential skills such as project management, communication, and interdisciplinary analysis, thereby preparing students for professional or academic success. Figure 1 provides a broad overview of how data moves through a system, highlighting the sequential steps of data engineering and the critical underlying disciplines that ensure its success.

Unlike *busy work* which often involves repetitive, low-cognitive tasks with limited connection to meaningful learning outcomes, the capstone project demands critical thinking, research, and collaboration. Here "reinventing the wheel" is not busy work but a deliberate method for gaining insights, building confidence, and preparing for real-world innovation. Drawing inspiration from Richard Feynman's quote, "What I cannot create, I do not understand", these are some of the great reasons to reinvent the wheel:

- Learn how wheels are made
- Teach others about wheels
- Learn about the inventors of wheels
- Be able to change wheels (or fix) when they break
- Build a better wheel (for some definition of better)
- Help someone in need of a very special wheel (maybe for a wheelchair?)
- Learn a tiny slice of what it means to build a larger system (such as a vehicle)

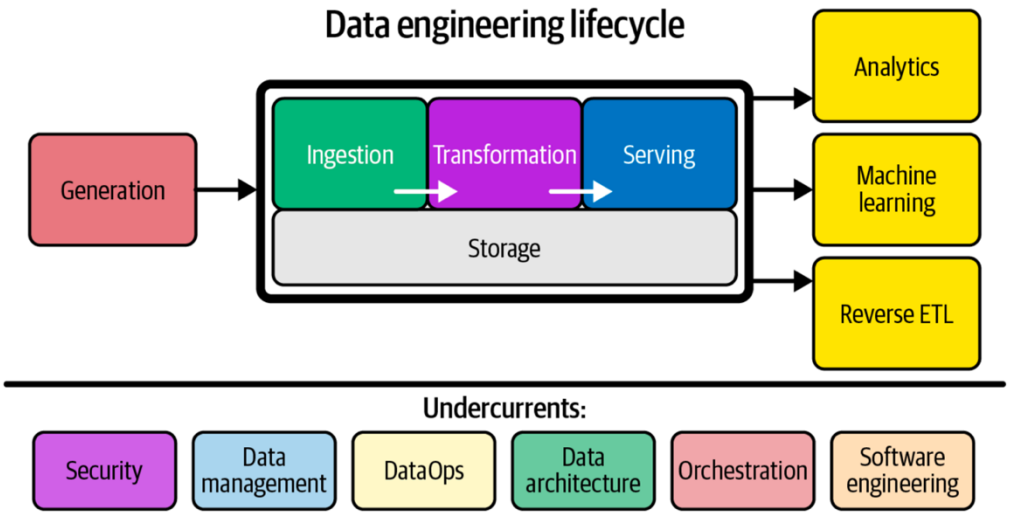


*Figure 1 – Components and undercurrents of the data engineering lifecycle. Image extracted from **Fundamentals of Data Engineering: Plan and Build Robust Data Systems**, J. Reis, M. Housley, 2022.*

V-model

The V-model (Fig. 2) provides a structured framework ideal for planning the Data Engineering Capstone Senior Design courses. It guides students from defining project goals and requirements to detailed system design and implementation. This model promotes a detailed and careful planning, iterative development, and continuous verification, helping students understand the full lifecycle of a data engineering system, from concept and design to deployment and maintenance. Each phase on the left is mirrored by corresponding testing and validation on the right, ensuring alignment between design and outcome.

Within this framework, a Project Charter is typically the first formal document, authorizing the work, defining objectives, scope, stakeholders, and approach, and giving each team the authority to access necessary resources and tools. Once the charter is approved, the Project Management Packet is assembled, detailing schedules, risks, communications, and other operational plans, and ensuring execution stays consistent with the V-model. Like a movie, the charter is the *green light*; the packet is the *production plan*. Below you have more details about the Project Charter and the Project Management Pack.

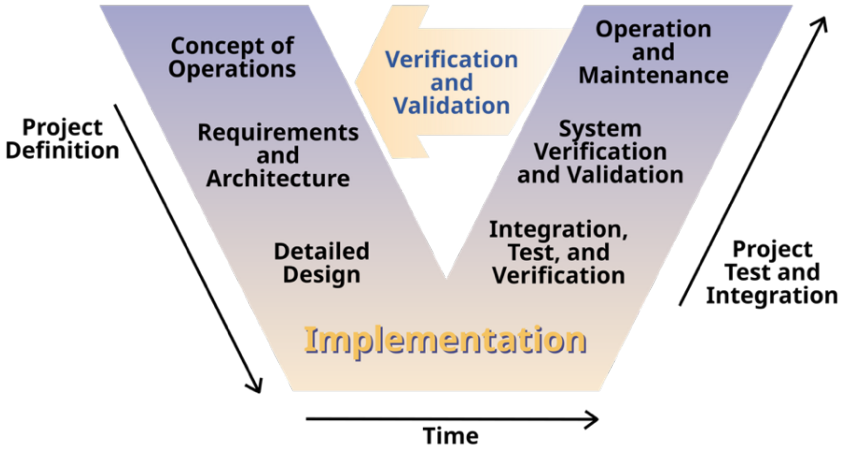


*Figure 2 – Systems Engineering Process. Image extracted from **Clarus: Concept of Operations, Federal Highway Administration (FHWA)**, 2005. (Publication No. FHWA-JPO-05-072, authors: L. Osborne, J. Brummond, R. Hart, M. Zarean, S. Conger.)*

1. Project Charter

The Project Charter is a concise, authoritative document that formally authorizes a project. In a two-semester capstone sequence, the final sponsor-signed charter is the *green light* to proceed to implementation in the second capstone semester.

Endorsed by the project sponsor, it grants the project manager and team the authority to commit organizational resources and establishes a common understanding of the project's purpose, scope, and constraints. Treat the charter as both a contract and a living reference; if conditions change, revise it formally rather than drifting informally. These are important aspects that need to be addressed in the charter:

1. **Authority & accountability**: Sponsor sign-off clarifies who owns vision and acceptance.
2. **Unified vision**: Forces the team to articulate a shared understanding before deep technical work.
3. **Baseline for changes**: Scope or resource changes must be justified against the charter, avoiding unpleasant surprises.
4. **Early risks**: Listing assumptions and constraints reveals potential (ethical/legal/technical) risks early enough to plan.
5. **Communication tool**: Because it is short, executives and non-technical stakeholders will actually read it.

2. Project Management Packet

The Project Management (PM) Packet is a single living document that guides a data-engineering capstone team from idea approval through final hand-off. Think of it as the project's constitution that captures what you will build, how, by whom, and when so faculty, sponsors, and teammates always have one authoritative source of truth. Keep the packet lean (about 10 pages) and version-controlled, and revisit it whenever scope, risks, or constraints change. A well-maintained packet not only streamlines your work, but it also showcases professional project-management discipline to future employers. How it's used:

1. **Proposal phase**: Submitted with your charter to secure faculty and sponsor sign-off.
2. **Execution phase**: Updated after each sprint review; serves as the agenda for weekly stand-ups.
3. **Evaluation phase**: Forms the backbone of your final report and memo, proving objectives were met.

Key components of the PM package and their functions:

- **Work Description**: Plain-language summary of what the team will build or analyze.
- **Deliverables**: Defines concrete outputs (code, datasets, report, poster) to hand over.
- **Project Boundaries**: States what is in and out to guard against scope creep.
- **Acceptance Criteria**: Minimum conditions the sponsor must see to sign off.
- **Constraints**: Non-negotiable limits (tools, data privacy, semester length, etc.).
- **Assumptions**: Conditions taken as true (e.g. data access) that may later change.
- **Quality and Performance Metrics**: How accuracy, reliability, and efficiency will be judged.
- **Metrics & Verification Methods**: How each metric will be measured, tested, and logged.
- **Timeline and Milestones**: Semester checkpoints that trigger progress reviews and pivots.
- **Task Priority**: Ranks tasks so critical-path items are tackled first.
- **Risks & Risk Management Plans**: Lists top pitfalls and the team's mitigation / contingency moves.
- **Cost Estimates and Budgeting**: Forecasts hours, cloud credits, and purchases to stay on budget.
- **Resources**: People, software, hardware, and data the project will rely on.

Initial guidelines – DAEN 460

I hope this simple comparison can help you understand how the two courses connect:

Dimension	DAEN 459	DAEN 460
Primary focus	Define and design	Implement and evaluate
Project Charter	Develop and approve	Maintain and verify compliance
PM Packet	Establish project plan	Execute, monitor, and update
Requirements	Elicit and specify	Verify and validate
Architecture	Design and justify	Implement and refine
Data	Identify and initially acquire	Integrate, process, and validate
Prototype	Demonstrate feasibility	Develop functional solution
Testing	Define evaluation strategy	Execute tests and document evidence
Sponsor engagement	Establish expectations	Validate results and acceptance
Final deliverable	Approved project proposal	Evaluated system and technical handoff

Note that, the goal of DAEN 460 is not simply to make something work, but to understand why it works, demonstrate how well it works, recognize when it does not work, and communicate what would be necessary to improve or maintain it.

Students should approach their capstone projects as engineers, not merely as users of technology. A successful project demonstrates the ability to translate requirements into a functional solution, make informed technical decisions, systematically evaluate results, and deliver trustworthy and reproducible engineering work.

One particularly important distinction for a Data Engineering capstone is that a dashboard, machine learning model, or analytical report may be a valuable output, but it should not automatically be considered sufficient evidence of data engineering competence. Students should also demonstrate meaningful work involving the underlying data architecture, integration, processing, quality, reliability, or delivery mechanisms.

1. From Design to Implementation

DAEN 460 represents the implementation and completion phase of the two-semester Data Engineering Capstone Senior Design experience. Building upon the project definition, requirements, system architecture, preliminary data acquisition, and initial prototype developed in DAEN 459, students are expected to transform their proposed solutions into functional, evaluated, and well-documented data engineering systems.

While DAEN 459 emphasizes understanding the problem and designing a feasible solution, DAEN 460 emphasizes executing the proposed design, overcoming implementation challenges, verifying technical performance, validating stakeholder requirements, and demonstrating measurable outcomes.

Students should recognize that a successful capstone project is not defined solely by whether a system produces an expected output. Success also depends on understanding how the system works, why specific engineering decisions were made, how well the system performs under realistic conditions, and whether the solution addresses the original problem. The goal is not simply to deliver a working product but to demonstrate the ability to apply engineering principles systematically throughout the entire development lifecycle.

2. Engineering Beyond Tools and Templates

Modern data engineering relies on an extensive ecosystem of frameworks, cloud services, libraries, and automated development tools. These technologies can substantially improve productivity, but their availability should not replace engineering understanding.

Students are encouraged to use appropriate existing technologies while demonstrating a thorough understanding of their underlying principles, limitations, and design trade-offs. The capstone experience should go beyond assembling preexisting components. Students must demonstrate meaningful technical contributions through activities such as designing data architectures, developing ingestion and transformation processes, implementing data quality controls, integrating heterogeneous systems, creating reusable software components, and evaluating alternative implementation approaches.

Students should be able to explain not only what tools they selected, but why those tools are appropriate and what would happen if the tools failed, were replaced, or encountered unexpected conditions.
The use of automated code generation and artificial intelligence tools do not eliminate the team's responsibility for understanding, verifying, testing, and documenting its implementation.

3. The V-Model: Verification and Validation

In DAEN 459, students primarily work through the left side of the V-model, defining requirements, establishing system architecture, and developing detailed designs. DAEN 460 places greater emphasis on implementation and the corresponding verification and validation activities represented on the right side of the V-model.

Students should distinguish between two fundamental engineering questions:

- **Verification**: Are we building the system, right? Verification determines whether implemented components and integrated systems satisfy their technical specifications and design requirements.
- **Validation**: Are we building the right system? Validation determines whether the completed solution addresses the sponsor's needs and intended use.

Every major requirement established in DAEN 459 should have an associated verification or validation method in DAEN 460. Depending on the project, these methods may include automated tests, performance benchmarks, data quality assessments, system demonstrations, comparative analyses, and sponsor acceptance reviews. Testing should not be treated as an activity performed only near the end of the semester. Instead, verification and validation should be integrated into development from the beginning.

4. End-to-End Data Engineering Thinking

Data engineering is fundamentally concerned with designing and maintaining reliable systems that transform raw data into trustworthy and usable information. Therefore, students should evaluate their projects from an end-to-end perspective rather than focusing exclusively on individual algorithms, scripts, dashboards, or software components.

Depending on project scope, relevant components may include data generation or acquisition, ingestion, storage, transformation, orchestration, data modeling, serving, monitoring, and downstream consumption. Students should demonstrate an understanding of how these components interact, where failures may occur, how data quality is preserved, and how the system could be maintained or extended. Not every project requires a large-scale cloud architecture or a production deployment. However, every project should demonstrate an appropriate level of engineering rigor relative to its goals, available resources, and technical complexity.

5. Data Quality, Reliability, and Reproducibility

A data engineering solution must be evaluated not only by its functionality but also by the trustworthiness of the data it processes and produces. Students should establish appropriate quality measures and testing procedures for their datasets, pipelines, and system outputs. Depending on the project, these measures may address completeness, accuracy, consistency, validity, uniqueness, timeliness, and integrity. Teams should also consider how their systems behave when confronted with missing values, duplicate records, unexpected data formats, unavailable sources, or other reasonably foreseeable failures.

Reproducibility is another essential engineering requirement. A technically competent person should be able to understand the system, identify its dependencies, reproduce its principal operations, and verify its reported results using the documentation and artifacts provided by the team. Students should maintain appropriate source code repositories, configuration files, environment specifications, testing procedures, and technical documentation throughout the semester.

6. Performance Evaluation and Engineering Trade-offs

A working solution is not necessarily an effective engineering solution. DAEN 460 students are expected to evaluate the performance of their systems using meaningful and measurable criteria established in relation to project objectives.
Depending on the project, evaluation may consider execution time, throughput, latency, scalability, data quality, resource utilization, availability, maintainability, analytical accuracy, or operational cost. Students should establish relevant performance baselines where feasible and use empirical evidence to assess improvements or limitations.

Engineering decisions frequently involve competing objectives. For example, improving system performance may increase cost, enhancing data security may affect usability, and increasing architectural complexity may reduce maintainability.
Students should identify significant trade-offs, explain their design decisions, and justify their recommendations using technical evidence rather than personal preferences or tool popularity.

7. Iterative Development and Project Management

Although the V-model provides a structured systems engineering framework, students should recognize that real-world engineering projects rarely progress through a perfectly linear sequence of activities. Unexpected data problems, changing requirements, technical limitations, and stakeholder feedback may require teams to revisit earlier design decisions.

DAEN 460 encourages an iterative approach in which teams implement, test, evaluate, and refine their solutions while maintaining alignment with the approved Project Charter. The Project Management Packet developed in DAEN 459 should remain an active management instrument rather than a document submitted once and forgotten.

Teams are expected to track progress against established milestones, document significant decisions, update risks, manage dependencies, and communicate deviations from the original plan. Changes to project scope, major deliverables, or acceptance criteria should be documented and reviewed with the appropriate faculty and sponsor representatives.

8. Professional Accountability and Team Contributions

Capstone projects are collaborative engineering efforts, and effective teamwork is an essential component of professional development. Each team member is expected to make identifiable technical and professional contributions to the project. Responsibilities should be clearly assigned, periodically reviewed, and adjusted when necessary.

Students should practice professional communication, constructive technical discussion, responsible decision-making, and timely reporting of project challenges. Collaboration does not mean that every team member must perform identical tasks. Instead, it requires that responsibilities are distributed appropriately, knowledge is shared, and individual contributions support the success of the complete system.

Every team member should understand the overall project architecture, the major design decisions, and how their work contributes to the final solution.

9. Sponsor Engagement and Acceptance

The project sponsor serves as a critical stakeholder throughout the capstone experience. Students should maintain regular communication with the sponsor to confirm expectations, review intermediate results, clarify requirements, and evaluate the usefulness of the proposed solution.

Sponsor feedback should be treated as an important source of validation, particularly when assessing whether the system addresses the intended operational or analytical problem. However, sponsor satisfaction alone does not establish technical quality. Students must also demonstrate that their solution meets appropriate engineering standards, evaluation criteria, and academic learning objectives. The final sponsor review should assess the completed deliverables against the approved acceptance criteria, including any formally agreed changes made during implementation.

10. Documentation, Knowledge Transfer, and Final Handoff

A capstone project is not complete simply because its code runs or its final presentation is delivered. Students must provide sufficient documentation and supporting materials for another technically qualified individual to understand the project's objectives, architecture, implementation, evaluation, and limitations.

The final project package should include the relevant technical artifacts, instructions for reproducing or operating the solution, results of verification and validation activities, and recommendations for future development.

Where appropriate, the handoff should also address dependencies, configuration requirements, operational constraints, known defects, and maintenance considerations.

The final report and presentation should communicate the engineering process as well as the outcome. Students should explain the original problem, proposed approach, implemented system, challenges encountered, alternative solutions considered, results obtained, and lessons learned. A successful handoff demonstrates that the team's work has value beyond the final course presentation.

11. Engineering Ethics and Responsible Data Use

Ethical responsibility remains an integral part of the engineering process throughout DAEN 460. Students must consider the consequences of their technical decisions, particularly when working with sensitive information, proprietary datasets, automated decision-making systems, or applications that may affect individuals or organizations. Relevant considerations may include data privacy, access control, security, intellectual property, fairness, transparency, appropriate data use, and compliance with applicable regulations and standards. Ethical and security considerations should influence the design, implementation, testing, and delivery of the solution rather than appearing solely as a brief discussion in the final report. Students are responsible for communicating significant limitations and uncertainties and should avoid presenting incomplete, unverified, or potentially misleading results as established conclusions.

12. Demonstrating Engineering Competence

The capstone experience should provide evidence that students can independently approach complex, unfamiliar, and imperfectly defined problems using the knowledge and skills developed throughout their undergraduate education. Students are expected to exercise engineering judgment, evaluate information critically, investigate alternatives, recognize limitations, and learn new technologies when necessary.

Technical difficulties, unsuccessful experiments, and revised approaches are natural components of engineering practice. These experiences can provide substantial educational value when students investigate their causes, document their findings, and apply the lessons to improve the solution.

The final evaluation should therefore consider not only the completed deliverables but also the quality of the engineering process, the depth of technical understanding, the justification of design decisions, and the ability to communicate results professionally. The overarching expectation is that students demonstrate ownership of their work and can confidently explain what they built, how they built it, why they built it that way, how they tested it, and what they learned.

13. From Deliverables to Evidence

Engineering claims must be supported by verifiable evidence. Students should not merely state that a system is reliable, efficient, scalable, accurate, secure, or successful. They must demonstrate these properties using appropriate evaluation methods and documented results. The following framework provides examples of how teams can demonstrate achievement of their project objectives. The amount and type of evidence should be proportional to the project's complexity and objectives. Students must distinguish between what has been implemented, what has been tested, what has been demonstrated, and what remains proposed or incomplete.

Engineering dimension	**Examples of expected evidence**
Requirements	Requirements traceability matrix and acceptance results
System implementation	Functional components, source code, architecture diagrams
Data quality	Validation tests, quality metrics, anomaly reports
Integration	End-to-end pipeline tests and integration results
Reliability	Failure-handling tests, recovery demonstrations
Performance	Benchmarks, resource measurements, baseline comparisons
Reproducibility	Environment specifications, execution instructions, repeatable results
Security and governance	Access controls, documented safeguards, compliance review
Project management	Updated PM Packet, milestone records, risk assessments
Sponsor acceptance	Demonstration, feedback, acceptance-criteria assessment
Knowledge transfer	Technical documentation, user instructions, final handoff